

Homework 2

Due Friday, February 6

1. Let Δ be a simplicial complex which is homeomorphic to M , a d -manifold with boundary ∂M . Prove that for $k \geq 0$,

$$(-1)^d(f_k(\Delta) - f_k(\partial\Delta)) = \sum_{j=k}^d (-1)^j \binom{j+1}{k+1} f_j(\Delta).$$

You may use that for any nonempty $\sigma \in \partial\Delta$, we have that $\text{lk}_\Delta \sigma$ is homotopy equivalent to a ball.

2. Find the h -vectors of the following simplicial complexes.

- (a) The n -simplex, $\Delta_n = 2^{[n]}$.
- (b) The boundary of the n -simplex, $\partial\Delta_n = 2^{[n]} \setminus \{[n]\}$.
- (c) The boundary of the cross polytope,

$$\partial C_n = \{\sigma \subset \{\pm 1, \dots, \pm n\} : \{i, -i\} \not\subset \sigma \text{ for all } i \in [n]\}.$$

- (d) The barycentric subdivision of $\partial\Delta_n$.

Hint: Use the Eulerian numbers.

3. (a) Prove that the convex hull of a compact set is compact.
Hint: Construct a continuous map $S \times \sigma \rightarrow \text{conv } S$, where σ is a simplex.
- (b) Prove that the conical hull of a compact set is not necessarily closed.
4. Let K be a closed, convex set. Prove that $v \in \text{lins } K$ if and only if there exists $x \in K$ such that $x + \lambda v \in K$ for all $\lambda \in \mathbb{R}$.